

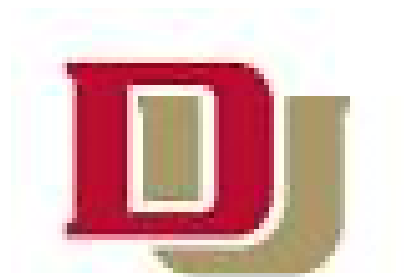
ENCE 3231 – Microprocessors 1

Final Project – Phase 1: System Evaluation

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Introduction

The purpose of this project is to evaluate the existing DemoSat High Altitude Balloon payload system and determine whether the current design is still feasible given component availability and project constraints. The DemoSat payload is a compact electronics system designed to collect atmospheric data during a high-altitude balloon flight. The system gathers environmental data using several sensors and stores the collected data on onboard storage for later analysis.

Phase 1 focuses on understanding the current system architecture, identifying all sensors and hardware components, estimating power consumption and weight, and determining whether the current design remains practical. This evaluation will help determine what improvements or modifications are required in Phase 2 of the project.

Current System Overview

The current DemoSat system is built around an Arduino Uno microcontroller, which acts as the main controller for the payload. The Arduino reads data from several environmental sensors, controls LED indicators, and stores collected data using a data logging module connected to a microSD card.

The system is powered by a 9V battery and includes a power switch that allows the payload to be turned on before launch.

The system currently includes the following components:

Controller

- Arduino Uno microcontroller

Sensors

- Internal temperature sensor
- External temperature sensor
- Humidity sensor
- Pressure sensor
- Accelerometer

Indicators

- LED indicators

Data Storage

- OpenLog data logging module
- microSD card

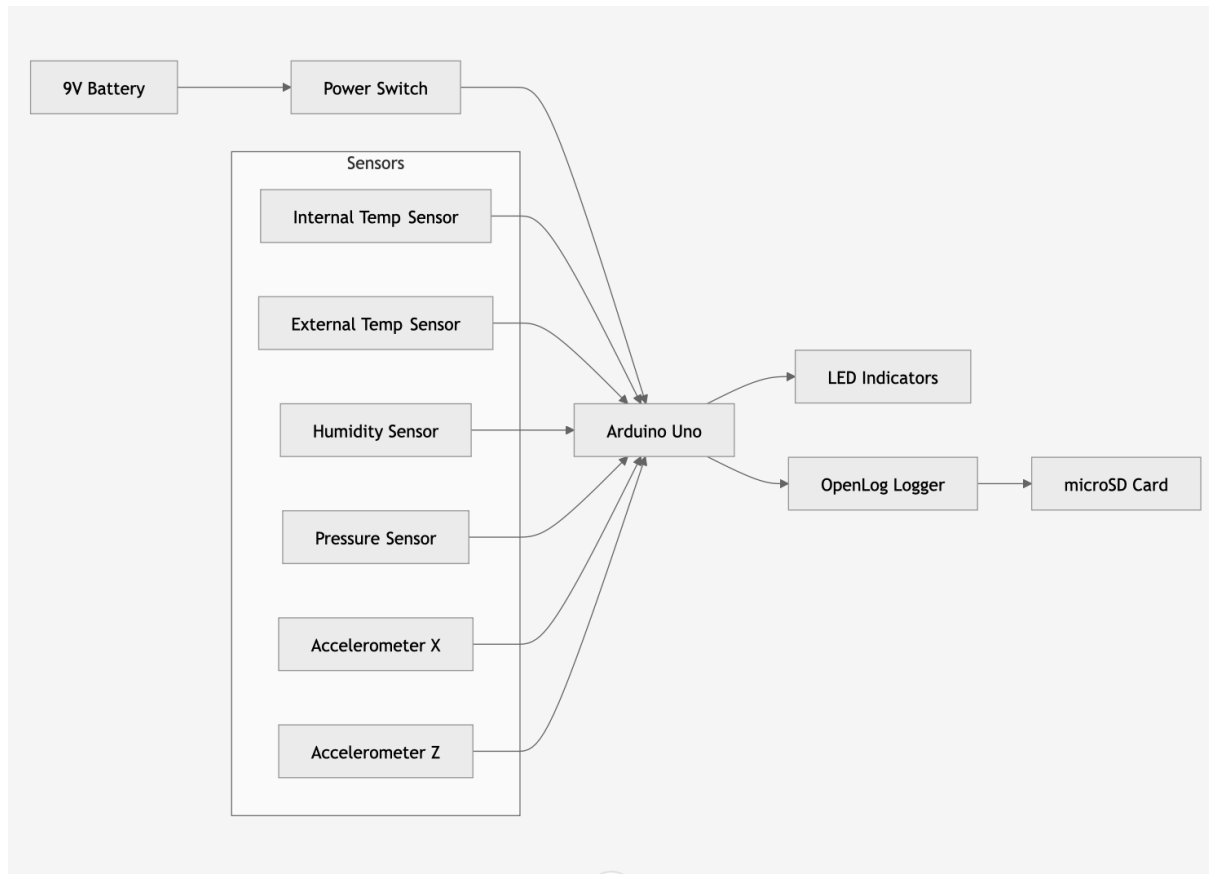
Power System

- 9V battery
- Power switch

These components allow the payload to measure atmospheric conditions such as temperature, humidity, pressure, and motion while storing the collected data for later retrieval.

System Block Diagram

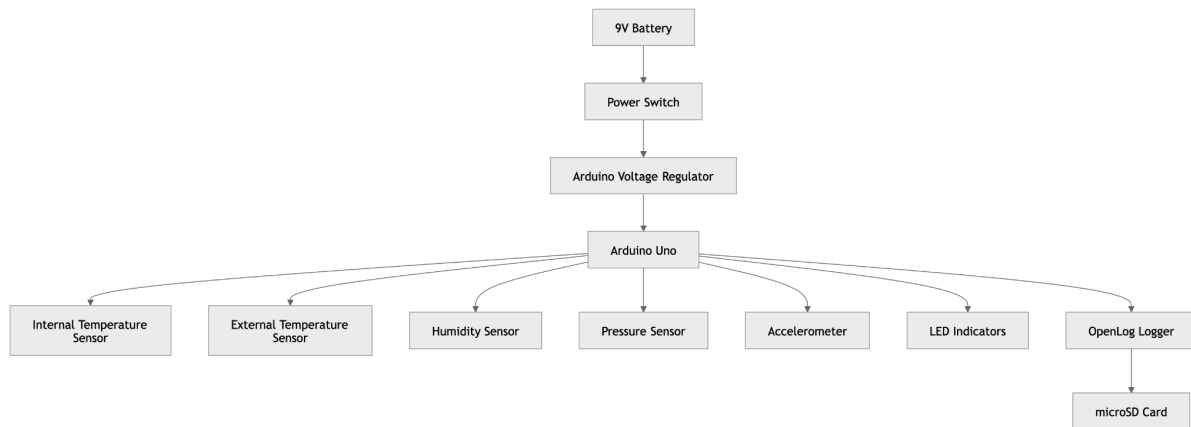
The following block diagram shows the architecture of the current DemoSat payload system. The block diagrams were originally sketched by hand and later formatted digitally using Mermaid for clarity and documentation.



In this system, the Arduino Uno reads sensor data through its analog input pins. The sensor data is processed by the microcontroller and then transmitted to the OpenLog module, which stores the information on a microSD card.

Figure 2 – Power Distribution Diagram

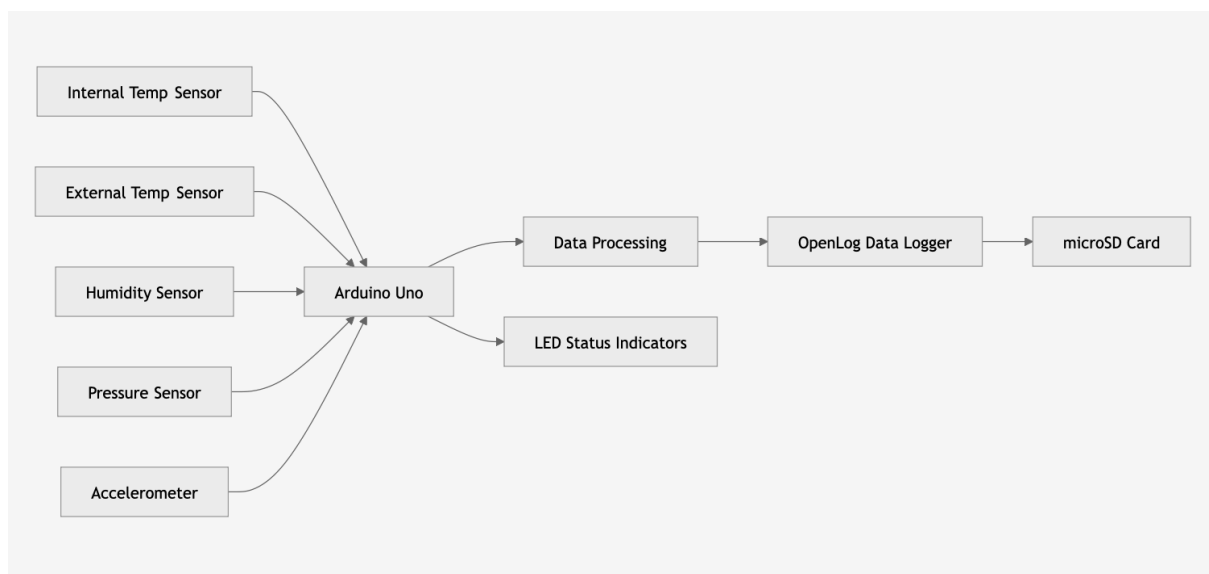
This diagram shows how electrical power flows from the battery through the system components.



This block diagram highlights the power distribution path, beginning with the 9V battery and flowing through the Arduino to power all connected components.

Figure 3 – Data Flow Diagram

This diagram illustrates how sensor data moves through the system and eventually gets stored.



This diagram focuses on the flow of information within the payload system. Sensor data is first collected by the Arduino, processed by the microcontroller, and then stored using the OpenLog module on the microSD card.

Estimated Power Consumption

To estimate the power consumption of the current system, the approximate current draw of each component can be considered.

Component:	Estimated Current:
Arduino Uno	50 mA
OpenLog Data Logger	15 mA
Pressure Sensor	7 mA
Humidity Sensor	0.2 mA
Accelerometer	0.4 mA

Temperature Sensors	0.1 mA
LED Indicators	10–20 mA

Estimated total current draw:

Total Current $\approx 85 - 90$ mA

Using the power equation:

$$P = V \times I$$

Assuming a system voltage of approximately 5 V:

$$P \approx 5 \times 0.09$$

$$P \approx 0.45 \text{ W}$$

Therefore, the estimated system power consumption is approximately:

0.4 W – 0.5 W

This level of power consumption is reasonable for a battery-powered payload system.

Estimated Electronics Weight

The electronics payload must remain under the maximum payload weight of 200 grams specified in the project requirements.

Estimated component weights are listed below:

Component	Estimated Weight
Arduino Uno	25 g
9V Battery	45 g
Pressure Sensor	4 g
OpenLog Module	1.8 g
Accelerometer	1.3 g
Humidity Sensor	1 g
Temperature Sensors	2 g
LEDs, wiring, headers, switch	10–15 g

Estimated total electronics weight:

Total $\approx$ 90 – 100 g

This value is well below the 200 g maximum payload weight requirement.

Feasibility Assessment

The existing DemoSat design successfully performs the required sensing and data logging tasks. It is capable of collecting temperature, humidity, pressure, and motion data during a balloon flight and storing this information on onboard storage.

However, several limitations exist in the current design.

One major issue is component availability. Some sensors originally included in the DemoSat sensor kit are no longer available for purchase.

This makes it difficult to reproduce the current system using the same parts.

Another limitation involves the accelerometer implementation. Although the accelerometer is capable of measuring motion along three axes, the current system only uses two analog inputs, meaning only two axes are currently measured.

The Arduino Uno also has limited analog input pins. Since the system already uses most of these inputs, it leaves little room for expansion or additional sensors.

Finally, the power system is not fully efficient. The system uses a 9V battery connected to the Arduino voltage regulator, which results in some power loss due to voltage regulation.

Phase 1 Conclusion

The current DemoSat payload system is capable of collecting environmental data and storing it during a high-altitude balloon flight. The estimated electronics weight is approximately 90–100 grams, which satisfies the maximum payload requirement of 200 grams. The estimated power consumption of approximately 0.4–0.5 watts is also reasonable for a battery-powered system.

Despite these strengths, the current design has limitations related to component availability, limited analog inputs, and incomplete use of the accelerometer's full capabilities. Because of these issues, a redesigned system will be proposed in Phase 2.

The next phase of the project will focus on selecting alternative components and designing a new payload system that maintains the required functionality while improving component availability, system flexibility, and power efficiency.